

POLITECNICO
MILANO 1863

SCUOLA DI INGEGNERIA INDUSTRIALE
E DELL'INFORMAZIONE

RASD – Requirement Analysis and Verification Document

Tesi di Laurea Magistrale in
Software Engineering 2

Author: **Alessandro Burato, Gabriele Crimella, Davide Faccini**

Student ID:
Advisor: Elisabetta Di Nitto
Co-advisors:
Academic Year: 2025-26

Contents

Contents **i**

1 Introduction **1**

1.1 Purpose 1

1.1.1 Goals 1

2 Overall Description **3**

3 Specific Requirements **5**

4 Formal Analysis Using Alloy **7**

5 Effort Spent **9**

6 References **11**

1 Introduction

1.1. Purpose

1.1.1. Goals

Here are the goals that BBP has to fulfill. BBP has to:

- G1: Recognize suitable bike paths
- G2: Allow the user to browse different bike paths
- G3: Allow the user to see bike paths statistics
- G4: Allow the user to search for paths between two locations
- G5: Allow the user to see paths optimality based on other registered users' reviews
- G6: Allow user to see weather information when available
- G7: Allow users to register to the application using credentials
- G8: Allow registered user to review published bike paths
- G9: Allow registered user to confirm or reject false positives in track recording
- G10: Allow registered user to let the system to automatically record suitable bike paths
- G11: Allow registered user to manually create suitable bike paths
- G12: Allow registered user to publish bike paths
- G13: Allow the store of bike paths data for registered users
- G14: Allow registered user to add publishable information about bike paths

2 | Overall Description

Chapter 2

3 | Specific Requirements

Chapter 3

4 | Formal Analysis Using Alloy

Chapter 4

5 | Effort Spent

Chapter 5

6 | References

Chapter 6

List of Figures

6.1 Caption of the Figure to appear in the List of Figures. 13
6.2 Shorter caption 13

List of Tables

6.1 Caption of the Table to appear in the List of Tables. 13
6.2 Highlighting the columns 14
6.3 Highlighting the rows 14